# JK BMS RS485 Modbus General Protocol (V1.0)

*Translated from Chinese Original — Technical/Literal Translation for Protocol Implementation*

The JK BMS RS485 Modbus General Protocol uses a master-slave request-response communication model. Only the master can initiate requests via the unique slave address, and the BMS (slave) responds according to the master's requests, implementing half-duplex communication. This protocol only allows the master to initiate requests while the slave responds passively; therefore, the slave will never actively occupy the communication line, preventing data collisions.

### Table of Contents


1. I. Physical Interface
2. II. Protocol Format
3. III. Error Feedback
4. IV. Information Transmission Process
5. V. Function Code Analysis
6. VI. Register Map — Configuration Registers (0x1000)
7. VII. Register Map — Real-Time Data Registers (0x1200)
8. VIII. Register Map — Device Information Registers (0x1400)
9. IX. Register Map — Command Registers (0x1600)


## I. Physical Interface

The electrical characteristics of the communication physical interface are as follows:

| Parameter | Value |
|---|---|
| Communication Interface | UART |
| Electrical Standard | RS485 |
| Baud Rate | 115200 bps |
| Data Bits | 8 |
| Stop Bits | 1 |
| Parity | None |

## II. Protocol Format

Information transmission is asynchronous, using hexadecimal for communication. The message frame format is:

| Address Code | Function Code | Data Area | CRC Checksum |
|---|---|---|---|
| 1 Byte | 1 Byte | N Bytes | 2 Bytes |

### 1) Address Code

The address code is the first byte of each communication frame. It supports values from 1 to 247. Each slave must have a unique address on the bus. Only the slave whose address matches the address code sent by the master will respond and return data.

### 2) Function Code

The function code is the second byte of each communication frame. Sent by the master, it instructs the slave device which operation to perform. The function code definitions are:

| Function Code | Definition | Operation |
|---|---|---|
| `03H` | Read Register(s) | Read data from one or more registers |
| `10H` | Write Register(s) | Write data to one or more registers |

### 3) Data Area

The data area varies depending on the function code and data direction. The data may be combinations such as "starting register address + number of registers to read", "register address + operation data", or "starting register address + number of registers to operate + data length + data", etc. Detailed data area formats for different function codes are explained in the "Function Code Analysis" section.

### 4) CRC Checksum

The CRC checksum is used to ensure the correctness and integrity of data transmission.

## III. Error Feedback

Address and CRC checksum errors will not receive any data feedback from the slave. Other errors will return error codes to the master. The second byte of the data frame is the original function code with `0x80` added, indicating a request error (illegal function code, illegal data value, etc.). The error data frame format is:

| Address Code | Function Code | Error Code Area | CRC Checksum |
|---|---|---|---|
| 1 Byte | 1 Byte | 1 Byte | 2 Bytes |

Error code definitions:

| Value | Name | Description |
|---|---|---|
| `01H` | Illegal Function Code | The function code is not supported for operating the register |
| `02H` | Register Address Error | Attempted to access a register that the slave prohibits access to |
| `03H` | Illegal Data | Data logic is invalid or exceeds limits |
| `04H` | CRC Checksum Error | CRC checksum error |

## IV. Information Transmission Process

When a communication command is sent by the master to the slave, the slave whose address matches the address code sent by the master receives the communication command. If the CRC checksum is correct, the slave executes the corresponding operation and returns the execution result (data) to the master. The return message contains the address code, function code, post-execution data, and CRC checksum. If the address does not match or the CRC checksum is incorrect, no information is returned.

## V. Function Code Analysis

### 1) Function Code 03H: Read Register(s)

**Example:** The master wants to read 2 holding registers starting from register address `05H` of slave address `01H`. Master sends:

| Master Sends | Data (HEX) |
|---|---|
| Address Code | `01H` |

| | | |
|---|---|---|
| Function Code | | 03H |
| Starting Register Address | High Byte | 00H |
| | Low Byte | 05H |
| Number of Registers | High Byte | 00H |
| | Low Byte | 02H |
| CRC Checksum | Low Byte | D4H |
| | High Byte | 0AH |

If the slave's holding registers **05H** and **06H** contain data **1122H** and **3344H**, the slave returns:

| Slave Returns | | Data (HEX) |
|---|---|---|
| Address Code | | 01H |
| Function Code | | 03H |
| Byte Count | | 04H |
| Register 05 Data | High Byte | 11H |
| | Low Byte | 22H |
| Register 06 Data | High Byte | 33H |
| | Low Byte | 44H |
| CRC Checksum | Low Byte | 4BH |
| | High Byte | C6H |

## 2) Function Code 10H: Write Register(s)

**Example:** The master wants to write data **0005H** and **2233H** to 2 registers starting at register address **0020H** of slave address **01H**. Master sends:

| Master Sends | | Data (HEX) |
|---|---|---|
| Address Code | | 01H |
| Function Code | | 10H |
| Starting Register Address | High Byte | 00H |
| | Low Byte | 20H |
| Number of Registers | High Byte | 00H |
| | Low Byte | 02H |
| Write Byte Count | | 04H |
| Register 0000H Data to Write | High Byte | 00H |
| | Low Byte | 05H |
| Register 0001H Data to Write | High Byte | 22H |
| | Low Byte | 33H |
| CRC Checksum | Low Byte | B9H |
| | High Byte | 03H |

Function code **10H** operation, slave returns:

| Slave Returns | Data (HEX) |
|---|---|
| Address Code | 01H |
| Function Code | 10H |

| | | |
|---|---|---|
| Starting Register Address | High Byte | **00H** |
| | Low Byte | **20H** |
| Number of Registers | High Byte | **00H** |
| | Low Byte | **02H** |
| CRC Checksum | Low Byte | **40H** |
| | High Byte | **02H** |

## VI. Register Map — Configuration Registers (Base Address: 0x1000)

These registers are used for BMS configuration parameters. All registers in this section are Read/Write (RW) with UINT32 type (4 bytes) unless otherwise specified.

### Protection & Voltage Configuration

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x0000 | 0 | UINT32 | 4 | RW | Smart Sleep Entry Voltage — VolSmartSleep | mV | |
| 0x0004 | 4 | UINT32 | 4 | RW | Cell Under-Voltage Protection — VolCellUV | mV | |
| 0x0008 | 8 | UINT32 | 4 | RW | Cell Under-Voltage Protection Recovery — VolCellUVPR | mV | |
| 0x000C | 12 | UINT32 | 4 | RW | Cell Over-Charge Protection — VolCellOV | mV | |
| 0x0010 | 16 | UINT32 | 4 | RW | Cell Over-Charge Protection Recovery Voltage — VolCellOVPR | mV | |
| 0x0014 | 20 | UINT32 | 4 | RW | Balance Trigger Voltage Difference — VolBalanTrig | mV | |
| 0x0018 | 24 | UINT32 | 4 | RW | SOC 100% Voltage — VolSOC100% | mV | |
| 0x001C | 28 | UINT32 | 4 | RW | SOC 0% Voltage — VolSOC0% | mV | |
| 0x0028 | 40 | UINT32 | 4 | RW | Auto Power-Off Voltage — VolSysPwrOff | mV | |

### Current Protection Configuration

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x002C | 44 | UINT32 | 4 | RW | Continuous Charge Current — CurBatCOC | mA | |
| 0x0030 | 48 | UINT32 | 4 | RW | Charge Over-Current Protection Delay — TIMBatCOCPDly | S | |
| 0x0034 | 52 | UINT32 | 4 | RW | Charge Over-Current Protection Release Delay — TIMBatCOCPRDly | S | |
| 0x0038 | 56 | UINT32 | 4 | RW | Continuous Discharge Current — CurBatDcOC | mA | |
| 0x003C | 60 | UINT32 | 4 | RW | Discharge Over-Current Protection Delay — TIMBatDcOCPDly | S | |
| 0x0040 | 64 | UINT32 | 4 | RW | Discharge Over-Current Protection Release Delay — TIMBatDcOCPRDly | S | |
| 0x0044 | 68 | UINT32 | 4 | RW | Short-Circuit Protection Release Delay — TIMBatSCPRDly | S | |
| 0x0048 | 72 | UINT32 | 4 | RW | Maximum Balancing Current — CurBalanMax | mA | |

### Temperature Protection Configuration

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x004C | 76 | INT32 | 4 | RW | Charge Over-Temperature Protection — TMPBatCOT | 0.1°C | |
| 0x0050 | 80 | INT32 | 4 | RW | Charge Over-Temperature Recovery — TMPBatCOTPR | 0.1°C | |
| 0x0054 | 84 | INT32 | 4 | RW | Discharge Over-Temperature Protection — TMPBatDcOT | 0.1°C | |
| 0x0058 | 88 | INT32 | 4 | RW | Discharge Over-Temperature Recovery — TMPBatDcOTPR | 0.1°C | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0x005C | 92 | INT32 | 4 | RW | Charge Under-Temperature Protection — TMPBatCUT | 0.1°C | |
| 0x0060 | 96 | INT32 | 4 | RW | Charge Under-Temperature Recovery — TMPBatCUTPR | 0.1°C | |
| 0x0064 | 100 | INT32 | 4 | RW | MOS Over-Temperature Protection — TMPMosOT | 0.1°C | |
| 0x0068 | 104 | INT32 | 4 | RW | MOS Over-Temperature Protection Recovery — TMPMosOTPR | 0.1°C | |

## Battery & Switch Configuration

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x006C | 108 | UINT32 | 4 | RW | Cell Count — CellCount | Series | |
| 0x0070 | 112 | UINT32 | 4 | RW | Charge Switch — BatChargeEN | | 1: ON; 0: OFF |
| 0x0074 | 116 | UINT32 | 4 | RW | Discharge Switch — BatDisChargeEN | | 1: ON; 0: OFF |
| 0x0078 | 120 | UINT32 | 4 | RW | Balance Switch — BalanEN | | 1: ON; 0: OFF |
| 0x007C | 124 | UINT32 | 4 | RW | Battery Design Capacity — CapBatCell | mAH | |
| 0x0080 | 128 | UINT32 | 4 | RW | Short-Circuit Protection Delay — SCPDelay | μs | |
| 0x0084 | 132 | UINT32 | 4 | RW | Balance Start Voltage — VolStartBalan | mV | |

## Connection Wire Internal Resistance (Cells 0–31)

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x0088 | 136 | UINT32 | 4 | RW | Connection Wire Resistance 0 — CellConWireRes0 | μΩ | |
| 0x008C | 140 | UINT32 | 4 | RW | Connection Wire Resistance 1 — CellConWireRes1 | μΩ | |
| 0x0090 | 144 | UINT32 | 4 | RW | Connection Wire Resistance 2 — CellConWireRes2 | μΩ | |
| 0x0094 | 148 | UINT32 | 4 | RW | Connection Wire Resistance 3 — CellConWireRes3 | μΩ | |
| 0x0098 | 152 | UINT32 | 4 | RW | Connection Wire Resistance 4 — CellConWireRes4 | μΩ | |
| 0x009C | 156 | UINT32 | 4 | RW | Connection Wire Resistance 5 — CellConWireRes5 | μΩ | |
| 0x00A0 | 160 | UINT32 | 4 | RW | Connection Wire Resistance 6 — CellConWireRes6 | μΩ | |
| 0x00A4 | 164 | UINT32 | 4 | RW | Connection Wire Resistance 7 — CellConWireRes7 | μΩ | |
| 0x00A8 | 168 | UINT32 | 4 | RW | Connection Wire Resistance 8 — CellConWireRes8 | μΩ | |
| 0x00AC | 172 | UINT32 | 4 | RW | Connection Wire Resistance 9 — CellConWireRes9 | μΩ | |
| 0x00B0 | 176 | UINT32 | 4 | RW | Connection Wire Resistance 10 — CellConWireRes10 | μΩ | |
| 0x00B4 | 180 | UINT32 | 4 | RW | Connection Wire Resistance 11 — CellConWireRes11 | μΩ | |
| 0x00B8 | 184 | UINT32 | 4 | RW | Connection Wire Resistance 12 — CellConWireRes12 | μΩ | |
| 0x00BC | 188 | UINT32 | 4 | RW | Connection Wire Resistance 13 — CellConWireRes13 | μΩ | |
| 0x00C0 | 192 | UINT32 | 4 | RW | Connection Wire Resistance 14 — CellConWireRes14 | μΩ | |
| 0x00C4 | 196 | UINT32 | 4 | RW | Connection Wire Resistance 15 — CellConWireRes15 | μΩ | |
| 0x00C8 | 200 | UINT32 | 4 | RW | Connection Wire Resistance 16 — CellConWireRes16 | μΩ | |
| 0x00CC | 204 | UINT32 | 4 | RW | Connection Wire Resistance 17 — CellConWireRes17 | μΩ | |
| 0x00D0 | 208 | UINT32 | 4 | RW | Connection Wire Resistance 18 — CellConWireRes18 | μΩ | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **0x00D4** | 212 | UINT32 | 4 | RW | Connection Wire Resistance 19 — CellConWireRes19 | μΩ | |
| **0x00D8** | 216 | UINT32 | 4 | RW | Connection Wire Resistance 20 — CellConWireRes20 | μΩ | |
| **0x00DC** | 220 | UINT32 | 4 | RW | Connection Wire Resistance 21 — CellConWireRes21 | μΩ | |
| **0x00E0** | 224 | UINT32 | 4 | RW | Connection Wire Resistance 22 — CellConWireRes22 | μΩ | |
| **0x00E4** | 228 | UINT32 | 4 | RW | Connection Wire Resistance 23 — CellConWireRes23 | μΩ | |
| **0x00E8** | 232 | UINT32 | 4 | RW | Connection Wire Resistance 24 — CellConWireRes24 | μΩ | |
| **0x00EC** | 236 | UINT32 | 4 | RW | Connection Wire Resistance 25 — CellConWireRes25 | μΩ | |
| **0x00F0** | 240 | UINT32 | 4 | RW | Connection Wire Resistance 26 — CellConWireRes26 | μΩ | |
| **0x00F4** | 244 | UINT32 | 4 | RW | Connection Wire Resistance 27 — CellConWireRes27 | μΩ | |
| **0x00F8** | 248 | UINT32 | 4 | RW | Connection Wire Resistance 28 — CellConWireRes28 | μΩ | |
| **0x00FC** | 252 | UINT32 | 4 | RW | Connection Wire Resistance 29 — CellConWireRes29 | μΩ | |
| **0x0100** | 256 | UINT32 | 4 | RW | Connection Wire Resistance 30 — CellConWireRes30 | μΩ | |
| **0x0104** | 260 | UINT32 | 4 | RW | Connection Wire Resistance 31 — CellConWireRes31 | μΩ | |

## Device & Miscellaneous Configuration

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| **0x0108** | 264 | UINT32 | 4 | RW | Device Address — DevAddr | H | |
| **0x010C** | 268 | UINT32 | 4 | RW | Discharge Pre-Charge Time — TIMProdischarge | S | |

## Bitfield Register at 0x0114 (UINT16, 2 bytes, RW)

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note / Bit |
|---|---|---|---|---|---|---|---|
| **0x0114** | 276 | UINT16 | 2 | RW | Heating Switch — HeatEN | | 1: ON; 0: OFF — BIT0 |
| | | | | RW | Temperature Sensor Disable — Disable temp-sensor | | 1: ON; 0: OFF — BIT1 |
| | | | | RW | GPS Heartbeat Detection — GPS Heartbeat | | 1: ON; 0: OFF — BIT2 |
| | | | | RW | Multiplexed Port Function — Port Switch | | 1: RS485; 0: CAN — BIT3 |
| | | | | RW | Display Always On — LCD Always On | | 1: ON; 0: OFF — BIT4 |
| | | | | RW | Dedicated Charger Recognition — Special Charger | | 1: ON; 0: OFF — BIT5 |
| | | | | RW | Smart Sleep — SmartSleep | | 1: ON; 0: OFF — BIT6 |

## Registers at 0x0116 and 0x0118

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| **0x0116** | 278 | INT8 | 2 | RW | Battery Alarm Temperature — TMPBatOTA | °C | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | INT8 | | RW | Battery Alarm Recovery Temperature — TMPBatOTAR | °C | |
| **0x0118** | 280 | UINT8 | 2 | RW | Smart Sleep Time — TIMSmartSleep | H | |
| | | UINT8 | | R | Data Domain Enable Control 0 | | |

## VII. Register Map — Real-Time Data Registers (Base Address: 0x1200)

These registers provide real-time battery monitoring data. All registers in this section are Read-Only (R).

### Individual Cell Voltages (Cells 0–31)

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x0000 | 0 | UINT16 | 2 | R | Cell Voltage 0 — CellVol0 | mV | |
| 0x0002 | 2 | UINT16 | 2 | R | Cell Voltage 1 — CellVol1 | mV | |
| 0x0004 | 4 | UINT16 | 2 | R | Cell Voltage 2 — CellVol2 | mV | |
| 0x0006 | 6 | UINT16 | 2 | R | Cell Voltage 3 — CellVol3 | mV | |
| 0x0008 | 8 | UINT16 | 2 | R | Cell Voltage 4 — CellVol4 | mV | |
| 0x000A | 10 | UINT16 | 2 | R | Cell Voltage 5 — CellVol5 | mV | |
| 0x000C | 12 | UINT16 | 2 | R | Cell Voltage 6 — CellVol6 | mV | |
| 0x000E | 14 | UINT16 | 2 | R | Cell Voltage 7 — CellVol7 | mV | |
| 0x0010 | 16 | UINT16 | 2 | R | Cell Voltage 8 — CellVol8 | mV | |
| 0x0012 | 18 | UINT16 | 2 | R | Cell Voltage 9 — CellVol9 | mV | |
| 0x0014 | 20 | UINT16 | 2 | R | Cell Voltage 10 — CellVol10 | mV | |
| 0x0016 | 22 | UINT16 | 2 | R | Cell Voltage 11 — CellVol11 | mV | |
| 0x0018 | 24 | UINT16 | 2 | R | Cell Voltage 12 — CellVol12 | mV | |
| 0x001A | 26 | UINT16 | 2 | R | Cell Voltage 13 — CellVol13 | mV | |
| 0x001C | 28 | UINT16 | 2 | R | Cell Voltage 14 — CellVol14 | mV | |
| 0x001E | 30 | UINT16 | 2 | R | Cell Voltage 15 — CellVol15 | mV | |
| 0x0020 | 32 | UINT16 | 2 | R | Cell Voltage 16 — CellVol16 | mV | |
| 0x0022 | 34 | UINT16 | 2 | R | Cell Voltage 17 — CellVol17 | mV | |
| 0x0024 | 36 | UINT16 | 2 | R | Cell Voltage 18 — CellVol18 | mV | |
| 0x0026 | 38 | UINT16 | 2 | R | Cell Voltage 19 — CellVol19 | mV | |
| 0x0028 | 40 | UINT16 | 2 | R | Cell Voltage 20 — CellVol20 | mV | |
| 0x002A | 42 | UINT16 | 2 | R | Cell Voltage 21 — CellVol21 | mV | |
| 0x002C | 44 | UINT16 | 2 | R | Cell Voltage 22 — CellVol22 | mV | |
| 0x002E | 46 | UINT16 | 2 | R | Cell Voltage 23 — CellVol23 | mV | |
| 0x0030 | 48 | UINT16 | 2 | R | Cell Voltage 24 — CellVol24 | mV | |
| 0x0032 | 50 | UINT16 | 2 | R | Cell Voltage 25 — CellVol25 | mV | |
| 0x0034 | 52 | UINT16 | 2 | R | Cell Voltage 26 — CellVol26 | mV | |
| 0x0036 | 54 | UINT16 | 2 | R | Cell Voltage 27 — CellVol27 | mV | |
| 0x0038 | 56 | UINT16 | 2 | R | Cell Voltage 28 — CellVol28 | mV | |
| 0x003A | 58 | UINT16 | 2 | R | Cell Voltage 29 — CellVol29 | mV | |
| 0x003C | 60 | UINT16 | 2 | R | Cell Voltage 30 — CellVol30 | mV | |
| 0x003E | 62 | UINT16 | 2 | R | Cell Voltage 31 — CellVol31 | mV | |

### Battery Status & Statistics

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x0040 | 64 | UINT32 | 4 | R | Cell Status — CellSta | | BIT[n]=1 indicates cell n is present |
| 0x0044 | 68 | UINT16 | 2 | R | Cell Average Voltage — CellVolAve | mV | |
| 0x0046 | 70 | UINT16 | 2 | R | Maximum Voltage Difference — CellVdifMax | mV | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0x0048 | 72 | UINT8 | 2 | R | Highest Voltage Cell Number — MaxVolCellNbr | | |
| | | UINT8 | | R | Lowest Voltage Cell Number — MinVolCellNbr | | |

## Balance Wire Resistance (Cells 0–31)

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x004A | 74 | UINT16 | 2 | R | Balance Wire Resistance 0 — CellWireRes0 | mΩ | |
| 0x004C | 76 | UINT16 | 2 | R | Balance Wire Resistance 1 — CellWireRes1 | mΩ | |
| 0x004E | 78 | UINT16 | 2 | R | Balance Wire Resistance 2 — CellWireRes2 | mΩ | |
| 0x0050 | 80 | UINT16 | 2 | R | Balance Wire Resistance 3 — CellWireRes3 | mΩ | |
| 0x0052 | 82 | UINT16 | 2 | R | Balance Wire Resistance 4 — CellWireRes4 | mΩ | |
| 0x0054 | 84 | UINT16 | 2 | R | Balance Wire Resistance 5 — CellWireRes5 | mΩ | |
| 0x0056 | 86 | UINT16 | 2 | R | Balance Wire Resistance 6 — CellWireRes6 | mΩ | |
| 0x0058 | 88 | UINT16 | 2 | R | Balance Wire Resistance 7 — CellWireRes7 | mΩ | |
| 0x005A | 90 | UINT16 | 2 | R | Balance Wire Resistance 8 — CellWireRes8 | mΩ | |
| 0x005C | 92 | UINT16 | 2 | R | Balance Wire Resistance 9 — CellWireRes9 | mΩ | |
| 0x005E | 94 | UINT16 | 2 | R | Balance Wire Resistance 10 — CellWireRes10 | mΩ | |
| 0x0060 | 96 | UINT16 | 2 | R | Balance Wire Resistance 11 — CellWireRes11 | mΩ | |
| 0x0062 | 98 | UINT16 | 2 | R | Balance Wire Resistance 12 — CellWireRes12 | mΩ | |
| 0x0064 | 100 | UINT16 | 2 | R | Balance Wire Resistance 13 — CellWireRes13 | mΩ | |
| 0x0066 | 102 | UINT16 | 2 | R | Balance Wire Resistance 14 — CellWireRes14 | mΩ | |
| 0x0068 | 104 | UINT16 | 2 | R | Balance Wire Resistance 15 — CellWireRes15 | mΩ | |
| 0x006A | 106 | UINT16 | 2 | R | Balance Wire Resistance 16 — CellWireRes16 | mΩ | |
| 0x006C | 108 | UINT16 | 2 | R | Balance Wire Resistance 17 — CellWireRes17 | mΩ | |
| 0x006E | 110 | UINT16 | 2 | R | Balance Wire Resistance 18 — CellWireRes18 | mΩ | |
| 0x0070 | 112 | UINT16 | 2 | R | Balance Wire Resistance 19 — CellWireRes19 | mΩ | |
| 0x0072 | 114 | UINT16 | 2 | R | Balance Wire Resistance 20 — CellWireRes20 | mΩ | |
| 0x0074 | 116 | UINT16 | 2 | R | Balance Wire Resistance 21 — CellWireRes21 | mΩ | |
| 0x0076 | 118 | UINT16 | 2 | R | Balance Wire Resistance 22 — CellWireRes22 | mΩ | |
| 0x0078 | 120 | UINT16 | 2 | R | Balance Wire Resistance 23 — CellWireRes23 | mΩ | |
| 0x007A | 122 | UINT16 | 2 | R | Balance Wire Resistance 24 — CellWireRes24 | mΩ | |
| 0x007C | 124 | UINT16 | 2 | R | Balance Wire Resistance 25 — CellWireRes25 | mΩ | |
| 0x007E | 126 | UINT16 | 2 | R | Balance Wire Resistance 26 — CellWireRes26 | mΩ | |
| 0x0080 | 128 | UINT16 | 2 | R | Balance Wire Resistance 27 — CellWireRes27 | mΩ | |
| 0x0082 | 130 | UINT16 | 2 | R | Balance Wire Resistance 28 — CellWireRes28 | mΩ | |
| 0x0084 | 132 | UINT16 | 2 | R | Balance Wire Resistance 29 — CellWireRes29 | mΩ | |
| 0x0086 | 134 | UINT16 | 2 | R | Balance Wire Resistance 30 — CellWireRes30 | mΩ | |
| 0x0088 | 136 | UINT16 | 2 | R | Balance Wire Resistance 31 — CellWireRes31 | mΩ | |

## Temperature, Voltage, Current & Power

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x008A | 138 | INT16 | 2 | R | Power Board Temperature — TempMos | 0.1°C | |
| 0x008C | 140 | UINT32 | 4 | R | Balance Wire Resistance Status — CellWireResSta | | BIT[n]=1 indicates balance wire n alarm |
| 0x0090 | 144 | UINT32 | 4 | R | Battery Total Voltage — BatVol | mV | |
| 0x0094 | 148 | UINT32 | 4 | R | Battery Power — BatWatt | mW | |
| 0x0098 | 152 | INT32 | 4 | R | Battery Current — BatCurrent | mA | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **0x009C** | 156 | INT16 | 2 | R | Battery Temperature 1 — TempBat 1 | 0.1°C | |
| **0x009E** | 158 | INT16 | 2 | R | Battery Temperature 2 — TempBat 2 | 0.1°C | |

## Alarm / Fault Status Register (0x00A0, UINT32, 4 bytes, R)

| Bit | Alarm Name (English) | Value Definition |
|---|---|---|
| BIT0 | Balance Wire Resistance Too High — AlarmWireRes | 1: Fault; 0: Normal |
| BIT1 | MOS Over-Temperature Protection — AlarmMosOTP | 1: Fault; 0: Normal |
| BIT2 | Cell Count Mismatch with Setting — AlarmCellQuantity | 1: Fault; 0: Normal |
| BIT3 | Current Sensor Abnormal — AlarmCurSensorErr | 1: Fault; 0: Normal |
| BIT4 | Cell Over-Voltage Protection — AlarmCellOVP | 1: Fault; 0: Normal |
| BIT5 | Battery Over-Voltage Protection — AlarmBatOVP | 1: Fault; 0: Normal |
| BIT6 | Charge Over-Current Protection — AlarmChOCP | 1: Fault; 0: Normal |
| BIT7 | Charge Short-Circuit Protection — AlarmChSCP | 1: Fault; 0: Normal |
| BIT8 | Charge Over-Temperature Protection — AlarmChOTP | 1: Fault; 0: Normal |
| BIT9 | Charge Under-Temperature Protection — AlarmChUTP | 1: Fault; 0: Normal |
| BIT10 | Internal Communication Abnormal — AlarmCPUAuxCommuErr | 1: Fault; 0: Normal |
| BIT11 | Cell Under-Voltage Protection — AlarmCellUVP | 1: Fault; 0: Normal |
| BIT12 | Battery Under-Voltage Protection — AlarmBatUVP | 1: Fault; 0: Normal |
| BIT13 | Discharge Over-Current Protection — AlarmDchOCP | 1: Fault; 0: Normal |
| BIT14 | Discharge Short-Circuit Protection — AlarmDchSCP | 1: Fault; 0: Normal |
| BIT15 | Discharge Over-Temperature Protection — AlarmDchOTP | 1: Fault; 0: Normal |
| BIT16 | Charge MOSFET Abnormal — AlarmChargeMOS | 1: Fault; 0: Normal |
| BIT17 | Discharge MOSFET Abnormal — AlarmDischargeMOS | 1: Fault; 0: Normal |
| BIT18 | GPS Disconnected — GPSDisconnected | 1: Fault; 0: Normal |
| BIT19 | Please Modify Authorization Password in Time — Modify PWD. in time | 1: Fault; 0: Normal |
| BIT20 | Discharge Enable Failed — Discharge On Failed | 1: Fault; 0: Normal |
| BIT21 | Battery Over-Temperature Alarm — Battery Over Temp Alarm | 1: Fault; 0: Normal |

## SOC, Balancing & Capacity Data

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x00A4 | 164 | INT16 | 2 | R | Balancing Current — BalanCurrent | mA | |
| 0x00A6 | 166 | UINT8 | 2 | R | Balancing Status — BalanSta | | 2: Discharging; 1: Charging; 0: Off |
| | | UINT8 | | R | State of Charge — SOCStateOfcharge | % | |
| 0x00A8 | 168 | INT32 | 4 | R | Remaining Capacity — SOCCapRemain | mAH | |
| 0x00AC | 172 | UINT32 | 4 | R | Battery Actual Full Charge Capacity — SOCFullChargeCap | mAH | |
| 0x00B0 | 176 | UINT32 | 4 | R | Cycle Count — SOCCycleCount | Times | |
| 0x00B4 | 180 | UINT32 | 4 | R | Cycle Total Capacity — SOCCycleCap | mAH | |
| 0x00B8 | 184 | UINT8 | 2 | R | State of Health Estimate — SOCSOH | % | |
| | | UINT8 | | R | Pre-Charge Status — Precharge | | 1: ON; 0: OFF |
| 0x00BA | 186 | UINT16 | 2 | R | User Layer Alarm — UserAlarm | | |
| 0x00BC | 188 | UINT32 | 4 | R | Run Time — RunTime | S | |
| 0x00C0 | 192 | UINT8 | 2 | R | Charge Status — Charge | | 1: ON; 0: OFF |
| | | UINT8 | | R | Discharge Status — Discharge | | 1: ON; 0: OFF |
| 0x00C2 | 194 | UINT16 | 2 | R | User Layer Alarm 2 — UserAlarm2 | | |

## Protection Release Timers

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x00C4 | 196 | UINT16 | 2 | R | Discharge Over-Current Protection Release Time — TimeDcOCPR | S | |
| 0x00C6 | 198 | UINT16 | 2 | R | Discharge Short-Circuit Protection Release Time — TimeDcSCPR | S | |
| 0x00C8 | 200 | UINT16 | 2 | R | Charge Over-Current Protection Release Time — TimeCOCPR | S | |
| 0x00CA | 202 | UINT16 | 2 | R | Charge Short-Circuit Protection Release Time — TimeCSCPR | S | |
| 0x00CC | 204 | UINT16 | 2 | R | Cell Under-Voltage Protection Release Time — TimeUVPR | S | |
| 0x00CE | 206 | UINT16 | 2 | R | Cell Over-Voltage Protection Release Time — TimeOVPR | S | |

## Temperature Sensor Status Register (0x00D0, UINT8, 2 bytes, R)

| Bit | Sensor Name | Value Definition |
|---|---|---|
| BIT0 | MOS Temperature Sensor — MOS TempSensorAbsent | 1: Present; 0: Absent |
| BIT1 | Battery Temperature Sensor 1 — BATTempSensor1Absent | 1: Present; 0: Absent |
| BIT2 | Battery Temperature Sensor 2 — BATTempSensor2Absent | 1: Present; 0: Absent |
| BIT4 | Battery Temperature Sensor 4 — BATTempSensor4Absent | 1: Present; 0: Absent |
| BIT5 | Battery Temperature Sensor 5 — BATTempSensor5Absent | 1: Present; 0: Absent |

**Note:** The second UINT8 byte at this address is: Heating Status — Heating (1: ON; 0: OFF)

## Miscellaneous Real-Time Data

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x00D2 | 210 | UINT16 | 2 | R | Reserved — Reserved | | |
| 0x00D4 | 212 | UINT16 | 2 | R | Emergency Switch Time — TimeEmergency | S | |
| 0x00D6 | 214 | UINT16 | 2 | R | Battery Current Correction Factor — BatCurCorrect | | |
| 0x00D8 | 216 | UINT16 | 2 | R | Charge Current Sensor Voltage — VolChargCur | mV | |
| 0x00DA | 218 | UINT16 | 2 | R | Discharge Current Sensor Voltage — VolDischargCur | mV | |
| 0x00DC | 220 | FLOAT | 4 | R | Battery Voltage Correction Factor — BatVolCorrect | | |
| 0x00E0 | 224 | UINT16 | 2 | R | Balance Charge PWM Duty Cycle — ChargePWMDutyCycle | % | |
| 0x00E2 | 226 | UINT16 | 2 | R | Balance Discharge PWM Duty Cycle — DischargePWMDutyCycle | % | |
| 0x00E4 | 228 | UINT16 | 2 | R | Battery Voltage — BatVol | 0.01V | |
| 0x00E6 | 230 | UINT16 | 2 | R | Heating Current — HeatCurrent | mA | |
| 0x00EE | 238 | UINT8 | 2 | R | Reserved — RVD | | |
| | | UINT8 | | R | Charger Status — ChargerPlugged | | 1: Plugged in; 0: Not plugged in |
| 0x00F0 | 240 | UINT32 | 4 | R | System Run Ticks — SysRunTicks | 0.1S | |
| 0x00F4 | 244 | UINT32 | 4 | R | PVD Trigger Timestamp — PVDTrigTimestamps | 0.1S | |
| 0x00F8 | 248 | INT16 | 2 | R | Battery Temperature 3 — TempBat 3 | 0.1°C | |
| 0x00FA | 250 | INT16 | 2 | R | Battery Temperature 4 — TempBat 4 | 0.1°C | |
| 0x00FC | 252 | INT16 | 2 | R | Battery Temperature 5 — TempBat 5 | 0.1°C | |
| 0x0100 | 256 | UINT32 | 4 | R | RTC Counter — RTCTicks | | Counting from 2020-01-01 |
| 0x0108 | 264 | UINT32 | 4 | R | Time to Enter Sleep — TimeEnterSleep | S | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **0x010C** | 268 | UINT8 | 2 | R | Parallel Current Limiting Module Status — PCLModuleSta | | 1: ON; 0: OFF |
| | | UINT8 | | | Reserved — RVD | | |

## VIII. Register Map — Device Information Registers (Base Address: 0x1400)

These registers provide read-only device identification and historical data.

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x0000 | 0 | ASCII | 16 | R | Manufacturer Device ID — ManufacturerDeviceID | | |
| 0x0010 | 16 | ASCII | 8 | R | Hardware Version — HardwareVersion | | |
| 0x0018 | 24 | ASCII | 8 | R | Software Version — SoftwareVersion | | |
| 0x0020 | 32 | UINT32 | 4 | R | Cumulative Run Time — ODDRunTime | S | |
| 0x0024 | 36 | UINT32 | 4 | R | Power-On Count — PWROnTimes | Times | |

## IX. Register Map — Command Registers (Base Address: 0x1600)

These registers accept Write-Only (W) commands for calibration, control, and battery chemistry presets.

| Address (HEX) | Offset (DEC) | Type | Len | R/W | Content (English) | Unit | Note |
|---|---|---|---|---|---|---|---|
| 0x0000 | 0 | UINT16 | 4 | W | Voltage Calibration — VoltageCalibration | mV | |
| 0x0004 | 4 | UINT16 | 2 | W | Protection Board Shutdown — Shutdown | | |
| 0x0006 | 6 | UINT16 | 4 | W | Current Calibration — CurrentCalibration | mA | |
| 0x000A | 10 | UINT16 | 2 | W | One-Key Ternary Lithium-Ion Preset — LI-ION | | |
| 0x000C | 12 | UINT16 | 2 | W | One-Key Lithium Iron Phosphate Preset — LiFePO4 | | |
| 0x000E | 14 | UINT16 | 2 | W | One-Key Lithium Titanate Preset — LTO | | |
| 0x0010 | 16 | UINT16 | 2 | W | Emergency Start — Emergency | | |
| 0x0012 | 18 | UINT32 | 4 | W | Time Calibration — TimeCalibration | | |

⚠ **Implementation Notes for Charger Integration:**

- All register offsets within a base address group are **byte offsets**, not Modbus register numbers. Divide byte offset by 2 to get the Modbus register address for 16-bit registers.
- Multi-byte values are transmitted in **Big-Endian** (high byte first) format as per standard Modbus convention.
- CRC checksum uses **CRC-16/Modbus** (polynomial 0xA001), transmitted low byte first.
- Temperature values in 0.1°C resolution: divide raw value by 10 to get degrees Celsius. Signed types (INT16/INT32) support negative temperatures.
- Current values (INT32 at 0x0098) are signed: positive = charge, negative = discharge.
- Always check the **Alarm Status Register (0x00A0)** before and during charging operations for safety-critical fault detection.

JK BMS RS485 Modbus General Protocol V1.0 — English Translation
Translated for charger-BMS communication protocol implementation